

1. A 42-year-old woman has the onset of chills 2 hours after transfusion of 2 units of packed red blood cells. She has a 1-week history of fatigue and shortness of breath with exertion; her hemoglobin concentration prior to the transfusion was 8.2 g/dL. Her temperature now is 38.5°C (101.3°F). The remainder of the examination shows no abnormalities. Her hemoglobin concentration now is 10 g/dL. Which of the following is the most likely explanation for this patient's fever?
- ☐ A) Bacterial contamination of the transfused blood
 - ☐ B) Hemolytic transfusion reaction
 - ☐ C) HIV infection from the transfused blood
 - ☐ D) IgA deficiency
 - ☐ E) Release of cytokines from the transfused blood



2. Twelve hours after admission to the hospital for evaluation of a right lower quadrant abdominal mass, a 77-year-old woman has dyspnea. On admission, hemoglobin concentration was 6 g/dL. CT scan of the abdomen showed a 10-cm complex mass of the cecum. The stool was brown and tested positive for occult blood. She has received 3 units of packed red blood cells since admission. Pulse is 120/min, respirations are 20/min, and blood pressure is 130/90 mm Hg. Pulse oximetry on 2 L/min of oxygen via nasal cannula shows an oxygen saturation of 92%. Crackles are heard bilaterally. Abdominal examination shows no tenderness. ECG shows sinus tachycardia. Which of the following is most likely to confirm the cause of this patient's respiratory distress?

- ☐ A) Arterial blood gas analysis
- ☐ B) Dipyridamole-thallium scan
- ☐ C) Post-transfusion crossmatch
- ☐ D) Ventilation-perfusion lung scans
- ☐ E) X-ray of the chest

3. A 36-year-old woman is admitted to the hospital because of bleeding in the upper gastrointestinal tract for 4 hours. Three months ago, she was hospitalized for 20 days for treatment of necrotizing pancreatitis secondary to choledocholithiasis. Four weeks ago, she was admitted for a similar episode of bleeding. Her temperature is 37.3°C (99.1°F), pulse is 110/min, and blood pressure is 110/70 mm Hg. The abdomen is soft with active bowel sounds. Endoscopy shows bleeding from gastric varices. The bleeding resolves with conservative treatment. Which of the following is the most likely cause of the bleeding?
- ☐ A) Alcoholic hepatitis
 - ☐ B) Cirrhosis of the liver
 - ☐ C) Fatty infiltration of the liver
 - ☐ D) Hepatic vein thrombosis
 - ☐ E) Splenic vein thrombosis

4. An 18-year-old man is brought to the emergency department 23 minutes after being involved in a motor vehicle collision in which he was thrown from the vehicle into a tree. At the scene, his spine was immobilized with a cervical collar and backboard, and intravenous administration of fluids was begun. On arrival, he is alert. His temperature is 35.6°C (96°F), pulse is 125/min, respirations are 30/min, and palpable systolic blood pressure is 60 mm Hg. Examination shows angulation of the midthoracic spine. He is unable to move his lower extremities. X-rays of the spine show compression fractures, angulation, and obliteration of the spinal canal at T7–8. After infusion of 3 L of lactated Ringer solution over the course of 20 minutes, his pulse is 90/min and he remains hypotensive. Which of the following is the most likely cause of this patient’s hypotension?
- ☐ A) Decreased myocardial contractility
 - ☐ B) Decreased sympathetic tone
 - ☐ C) Hypothermia
 - ☐ D) Hypovolemia
 - ☐ E) Increased capillary permeability



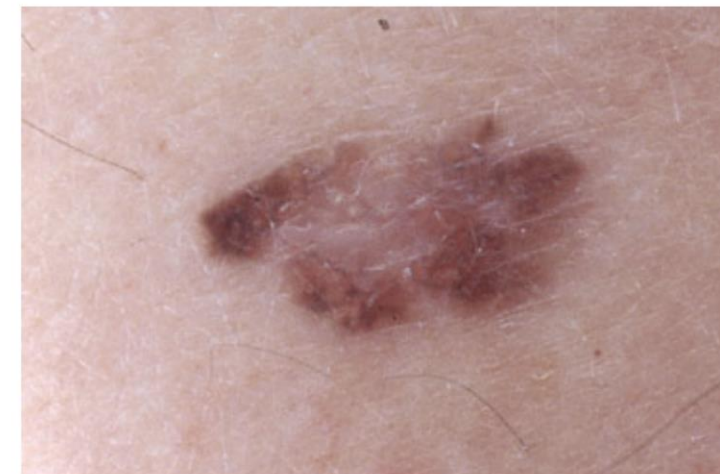
5. A 67-year-old man comes to the physician because of low back pain for 1 week and urinary incontinence for 2 days. He is currently undergoing chemotherapy for squamous cell carcinoma of the lung with metastases to the liver. Examination shows tenderness to palpation over the lumbar spine and decreased sensation below the umbilicus. Rectal examination shows decreased sphincter tone. The prostate is diffusely enlarged. Which of the following is the most appropriate next step in diagnosis?

- ☐ A) Measurement of serum prostate-specific antigen concentration
- ☐ B) Urine culture
- ☐ C) Cystourethrography
- ☐ D) Renal ultrasonography
- ☐ E) CT scan of the head
- ☐ F) MRI of the spine

6. A 25-year-old woman is brought to the emergency department on a backboard with her neck and head restrained after a motor vehicle collision; she was not wearing a seat belt. She is awake but confused and has no memory of the crash. Examination shows a contusion on the right side of the head and clear fluid draining from the right ear. Which of the following is the most likely explanation for these findings?
- ☐ A) Comminuted skull fracture causing lymphatic disruption in the ear
 - ☐ B) Cranial fracture that has disrupted the meninges
 - ☐ C) Linear fracture of the temporomandibular joint
 - ☐ D) Stellate fracture of the frontoparietal bone
 - ☐ E) Zygomatic fracture

7. A 67-year-old woman comes to the physician because of a painless large freckle on the back of her thigh that she first noticed 1 month ago. She has no history of serious illness and takes no medications. There is no personal or family history of skin lesions. She uses sunscreen with SPF 45 regularly. The patient is fair skinned. Examination shows a 1-cm lesion above the popliteal fossa. A photograph of the lesion is shown. Which of the following is the most likely diagnosis?

- ☐ A) Benign nevus
- ☐ B) Keratoacanthoma
- ☐ C) Melanoma
- ☐ D) Seborrheic keratosis
- ☐ E) Solar lentigo



8. A 65-year-old man comes for a follow-up examination 1 week after excision of a small raised lesion with pearly margins from the lateral left naris. The pathology confirms basal cell carcinoma with the tumor extending to the resection margin in one section. Examination shows a well-healed incision. Which of the following is the most appropriate next step in management?
- ☐ A) Observation and reexamination in 6 months
 - ☐ B) Laser therapy to the incision
 - ☐ C) Radiation therapy to the nose
 - ☐ D) Reexcision of the excision site
 - ☐ E) Cervical lymph node dissection



9. A 57-year-old man is brought to the emergency department 30 minutes after sustaining burns over 40% of his total body surface area in a house fire. His pulse is 110/min, respirations are 22/min, and blood pressure is 120/70 mm Hg. The burned surface is anesthetic and covered with a thick, firm, brownish eschar; thrombosed vessels are visible through the burn surface. Over the next 24 hours, which of the following is the most urgent priority in management for this patient?

- ☐ A) Hyperbaric oxygen therapy
- ☐ B) Administration of tetanus toxoid
- ☐ C) Intravenous fluid resuscitation
- ☐ D) Broad-spectrum antibiotic therapy
- ☐ E) Excision and grafting



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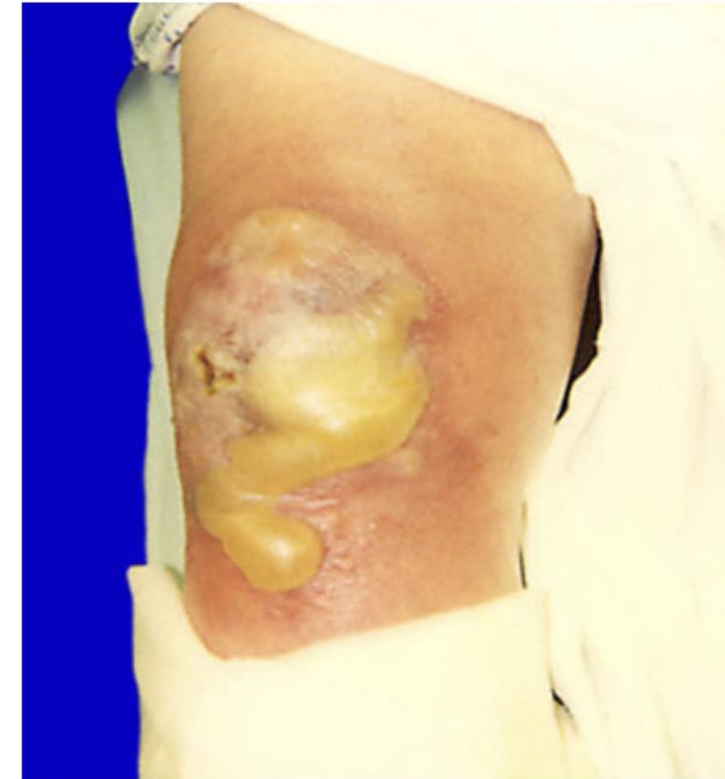


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10. A 47-year-old man comes to the emergency department 3 days after he sustained a painful injury to his right knee in a fall. He explains that he thought he had only scraped his knee but that the wound has been getting progressively worse. He has a 20-year history of type 2 diabetes mellitus well controlled with insulin. His most recent tetanus vaccination was 3 years ago. His temperature is 38.4°C (101.1°F), pulse is 100/min and regular, respirations are 12/min, and blood pressure is 155/60 mm Hg. A photograph of the right knee is shown. Palpation of the affected knee produces pain and crepitation. An x-ray of the right knee shows gas in the soft tissue and no bony abnormalities. Analgesic therapy is begun, and urgent surgical debridement is planned. Which of the following is the most appropriate next step in management?

- ☐ A) MRI of the knee
- ☐ B) Application of silver sulfadiazine
- ☐ C) Antibiotic therapy
- ☐ D) Corticosteroid therapy
- ☐ E) Administration of tetanus toxoid
- ☐ F) Hyperbaric oxygen therapy



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11. A previously healthy 19-year-old woman is brought to the emergency department 45 minutes after sustaining an injury to her upper arm while playing ice hockey. Examination shows diffuse swelling over the upper arm; the skin is intact. There is absence of wrist extension. X-rays show a fracture of the distal humerus. Which of the following is the most likely site of nerve injury?
- ☐ A) Axillary nerve
 - ☐ B) Median nerve above the elbow
 - ☐ C) Median nerve at the elbow
 - ☐ D) Musculocutaneous nerve above the elbow
 - ☐ E) Radial nerve above the elbow
 - ☐ F) Radial nerve at the elbow
 - ☐ G) Ulnar nerve above the elbow
 - ☐ H) Ulnar nerve at the elbow

12. An 82-year-old man is admitted to the intensive care unit after undergoing an uncomplicated, elective sigmoid colectomy for chronic diverticulitis. He is extubated and rates his pain as a 2 on a 10-point scale. He has a history of coronary artery disease. His medications are atorvastatin, metoprolol, and aspirin. His temperature is 37.2°C (99°F), pulse is 110/min, respirations are 32/min, and blood pressure is 140/80 mm Hg. Pulse oximetry on 6 L/min of oxygen by nasal cannula shows an oxygen saturation of 92%. There is jugular venous distention. Expiratory wheezes are heard at both lung bases. Which of the following is the most likely diagnosis?
- ☐ A) Congestive heart failure
 - ☐ B) Cor pulmonale
 - ☐ C) Myocardial infarction
 - ☐ D) Pneumonia
 - ☐ E) Pulmonary embolism

13. A 30-year-old woman has had frequent palpitations and dyspnea with mild exertion for 2 days. She underwent an aortic valve replacement for rheumatic heart disease 2 years ago. She is receiving warfarin therapy. Her pulse is 80/min and regular, and blood pressure is 140/palpable mm Hg. She has pale conjunctivae. The lungs are clear to auscultation. Prosthetic heart sounds and loud systolic and diastolic murmurs are heard. There is trace edema of the lower extremities. Hematocrit is 24% with numerous schistocytes and fractured erythrocytes. Leukocyte count is 14,000/mm³. An x-ray of the chest shows cardiomegaly without redistribution of blood flow. Which of the following is the most likely cause of this patient's condition?
- ☐ A) Cor pulmonale
 - ☐ B) Dysfunction of the prosthetic valve
 - ☐ C) Pulmonary emboli
 - ☐ D) Ruptured papillary muscle
 - ☐ E) Warfarin-induced hemolytic anemia



14. One day after undergoing open cholecystectomy, a 47-year-old woman has increased blood pressure and moderate abdominal pain. She was not taking any medications prior to the operation. Postoperatively, she has been receiving lactated Ringer solution (125 mL/h) and patient-controlled morphine with a bolus dose of 1 mg and a maximum dose of 6 mg/h; there is a 10-minute lockout interval and no basal rate. She has been given nothing orally. She is alert and cooperative. Temperature is 37.0°C (98.6°F), pulse is 115/min, respirations are 22/min and shallow, and blood pressure is 170/90 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 95%. There is no jugular venous distention or edema. Cardiopulmonary examination is normal. The abdomen is soft with moderate tenderness; movement produces pain. Bowel sounds are absent. Which of the following is the most appropriate next step in management of this patient's hypertension?
- ☐ A) Increase bolus of morphine
 - ☐ B) Increase rate of lactated Ringer solution
 - ☐ C) Intravenous furosemide therapy
 - ☐ D) Intravenous metoprolol therapy
 - ☐ E) Oral hydrochlorothiazide therapy
 - ☐ F) Oral nifedipine therapy

15. A 67-year-old man comes to the physician for a routine health maintenance examination. He feels well. He has hypertension well controlled with enalapril. Five years ago, he underwent coronary artery bypass grafting. He smoked two packs of cigarettes daily for 20 years but stopped smoking 20 years ago. He walks 1 mile daily. His temperature is 37°C (98.6°F), pulse is 84/min, respirations are 16/min, and blood pressure is 125/78 mm Hg. Cardiopulmonary examination shows no abnormalities. Abdominal examination shows a nontender pulsatile mass in the mid abdomen; no bruits are heard. Femoral pulses are normal. Abdominal ultrasonography shows a 4-cm aneurysm; there is no evidence of rupture. Which of the following factors provides the best rationale for serial imaging rather than surgical repair in this patient?
- ☐ A) Absence of symptoms
 - ☐ B) Age of the patient
 - ☐ C) The patient's history of cardiac disease
 - ☐ D) The patient's history of hypertension
 - ☐ E) Size of the aneurysm

16. A 30-year-old male firefighter has been hospitalized for 3 weeks for treatment of third-degree burns involving the neck, upper anterior chest, and right upper extremity. A purulent exudate is noted at an intravenous catheter site located in the proximal cephalic vein just above the left wrist. When the catheter is removed, additional purulent exudate drains from the site. Temperature is 38.7°C (101.7°F), pulse is 115/min, respirations are 18/min, and blood pressure is 105/70 mm Hg. Physical examination shows an 8-cm segment of the cephalic vein that is firm, tender, indurated, erythematous, and fluctuant. In addition to intravenous broad-spectrum antibiotics, which of the following is the most appropriate next step in management?
- ☐ A) Application of a negative pressure dressing
 - ☐ B) Elevation of the left arm and application of warm compresses
 - ☐ C) Excision of the vein
 - ☐ D) Intravenous administration of heparin
 - ☐ E) Intravenous administration of tissue plasminogen activator

17. A 58-year-old woman has had pain and tenderness of the long bones of the lower extremities, knees, and ankles for 2 months. She has smoked one pack of cigarettes daily for 30 years. She was diagnosed with squamous cell carcinoma of the lung 6 months ago. She has had rheumatoid arthritis for 15 years. She fractured her ankle in a fall 6 years ago. Examination of the lower extremities shows tenderness to palpation of the femur and tibia bilaterally and slight warmth and tenderness of the knees and ankles. X-rays show periosteal new bone formation of the femurs. Her symptoms are most likely a complication of which of the following?

- ☐ A) Osteoporosis
- ☐ B) Primary hyperparathyroidism
- ☐ C) Prior trauma to the ankle
- ☐ D) Rheumatoid arthritis
- ☐ E) Squamous cell carcinoma of the lung



18. Six days after abdominoperineal resection of the rectum for cancer, a 62-year-old woman becomes dyspneic as she walks to the bathroom. The postoperative course had been uncomplicated. Her temperature is 37.1°C (98.7°F), pulse is 135/min, respirations are 34/min, and blood pressure is 120/70 mm Hg. She is anxious and diaphoretic. The chest is clear to auscultation and there are no cardiac murmurs or gallops. There is no tenderness or edema of the lower extremities. Which of the following is the most likely cause?

- ☐ A) Atelectasis
- ☐ B) Congestive heart failure
- ☐ C) Myocardial infarction
- ☐ D) Pneumonia
- ☐ E) Pulmonary embolism



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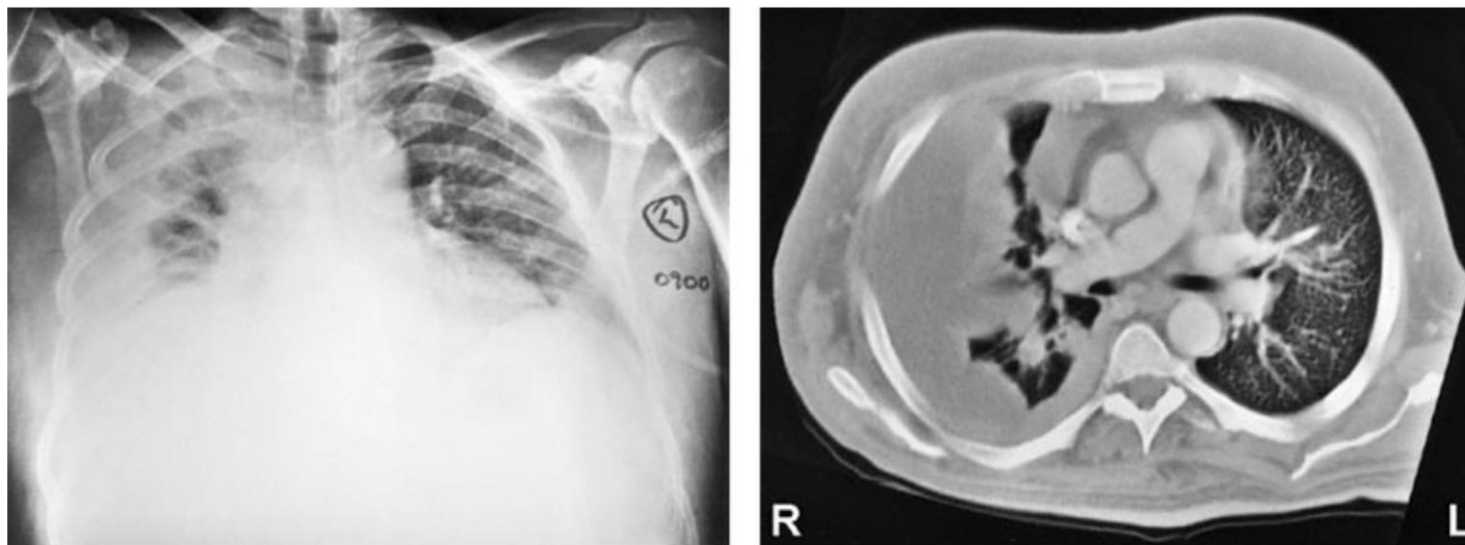
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19. Two months after undergoing an orthotopic liver transplant, a 47-year-old man comes to the physician because of persistent fever and shortness of breath. He has had recurrent pleural effusions since the operation requiring aspiration; the yield of fluid has been decreasing each time. His temperature is 38°C (100.4°F), and respirations are 24/min. Breath sounds are decreased over the right hemithorax, and there is dullness to percussion. The remainder of the examination shows no abnormalities. An x-ray and a CT scan of the chest are shown. Which of the following is the most likely diagnosis?

- ☐ A) Chronic atelectasis
- ☐ B) Empyema
- ☐ C) Lung abscess
- ☐ D) Mesothelioma
- ☐ E) Persistent pneumothorax

20. A 16-year-old boy is brought to the emergency department after a motor vehicle collision; he was not wearing a seat belt and struck his chest on the dashboard. Examination shows a fracture of the right arm; an x-ray of the chest shows minimal haziness in the right midlung field. Forty-eight hours later, he has a cough, and crackles are heard on auscultation. He is afebrile. A repeat x-ray shows confluent infiltrates over the left and right upper lobes. Which of the following is the most likely explanation for these findings?

- ☐ A) Anaerobic pneumonitis
- ☐ B) Aspiration pneumonia
- ☐ C) Fat embolism
- ☐ D) Lung contusion
- ☐ E) Thromboembolism

21. An obese 37-year-old man is brought to the emergency department 30 minutes after being involved in a motor vehicle collision. He was the restrained front-seat passenger. His pulse is 90/min, respirations are 24/min, and blood pressure is 110/70 mm Hg. Ecchymoses are visible over the midabdomen from the lap belt. A nasogastric tube is inserted. An x-ray of the chest shows an obscured left hemidiaphragm with the tip of the nasogastric tube in the lower left side of the chest. Which of the following is the most likely diagnosis?

- ☐ A) Diaphragmatic rupture
- ☐ B) Esophageal rupture
- ☐ C) Pancreatic injury
- ☐ D) Pulmonary contusion
- ☐ E) Ruptured spleen

22. A 47-year-old man is admitted to the hospital for evaluation of a 7-day history of fever and diffuse mild abdominal pain. During the past 2 months, he has noticed increased abdominal girth. He has a long-standing history of alcoholism. His temperature is 38.5°C (101.3°F), pulse is 105/min, respirations are 16/min, and blood pressure is 135/82 mm Hg. Examination shows poor inspiratory effort. Heart sounds are normal. The abdomen is distended with mild tenderness throughout all quadrants and shifting dullness. Bowel sounds are decreased. Rectal examination shows no abnormalities. Paracentesis is performed. Laboratory studies show:

Serum	
Protein, total	5.2 g/dL
Albumin	3.4 g/dL
Globulin	1.8 g/dL
Peritoneal fluid	
Leukocyte count	3600/mm ³
Segmented neutrophils	68%
Monocytes	18%
Protein, total	1.1 g/dL
Albumin	0.7 g/dL

Which of the following is the most likely diagnosis?

- ☐ A) Alcoholic hepatitis
- ☐ B) Carcinomatosis
- ☐ C) Enteric perforation
- ☐ D) Peritoneal tuberculosis
- ☐ E) Spontaneous bacterial peritonitis

23. A 27-year-old man comes to the physician because of a 6-month history of intermittent abdominal pain and a decreased energy level. The first episode of abdominal pain occurred after he ate a large meal. He then had an urgent need to move his bowels and passed watery stool containing bright red blood. At first, subsequent episodes occurred once or twice weekly and only involved abdominal pain. Four weeks ago, he started to have daily pain associated with bloody, watery stools. He has no history of serious illness and takes no medications. He visited Mexico 1 year ago. He is sexually active with one male partner, and they do not use condoms consistently. He appears pale and tired. He is 191 cm (6 ft 3 in) tall and weighs 83 kg (183 lb); BMI is 23 kg/m². His temperature is 36.9°C (98.4°F), pulse is 85/min, respirations are 18/min, and blood pressure is 115/70 mm Hg. The abdomen is scaphoid with mild, diffuse tenderness. No masses are palpated. Laboratory studies show:

Hemoglobin	9.3 g/dL
Hematocrit	29%
Leukocyte count	7800/mm ³
Platelet count	310,000/mm ³
Serum	
Antineutrophil cytoplasmic antibody	positive
<i>Saccharomyces cerevisiae</i> antibody	negative

Flexible sigmoidoscopy shows pseudopolyps and inflamed, friable mucosa. Which of the following is the most likely diagnosis?

- ☐ A) Amebic dysentery
- ☐ B) Crohn disease
- ☐ C) Cytomegalovirus proctitis
- ☐ D) Ulcerative colitis
- ☐ E) *Yersinia enterocolitis*

24. A 55-year-old woman has had vague pain in the left lower quadrant of the abdomen for 4 months and foul-smelling urine for 2 weeks. She appears thin. There is mild tenderness in the left lower quadrant of the abdomen. Rectal examination shows tenderness with no masses. Hematocrit is 29%. Urinalysis shows 50–100 WBC/hpf and 4+ bacteria. Urine culture grows greater than 100,000 colonies/mL of mixed fecal flora. Test of the stool for occult blood is positive. Which of the following is the most likely underlying diagnosis?
- ☐ A) Amebiasis
 - ☐ B) Colon cancer
 - ☐ C) Familial polyposis
 - ☐ D) Intussusception
 - ☐ E) Ulcerative colitis



25. A 68-year-old man has had an enlarging lump under his tongue over the past 3 months. The lesion is now ulcerated and somewhat painful. He quit smoking cigarettes 10 years ago. There is a mass adjacent to the inferior dental arch extending to the inferior surface of the tongue. Submental lymph nodes are palpable, but there is no neck adenopathy. Which of the following is the most appropriate next step in diagnosis?

- ☐ A) Panoramic x-ray of the jaw
- ☐ B) Barium swallow
- ☐ C) Biopsy of the mass
- ☐ D) Biopsy of the lymph nodes
- ☐ E) Wide excision of the mass with radical neck dissection



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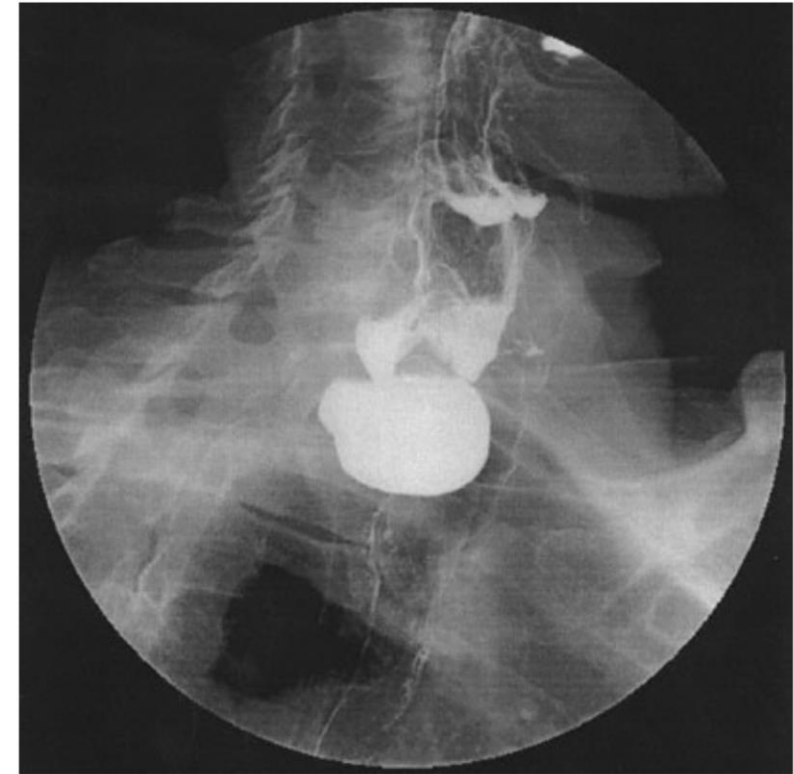


26. A 67-year-old man remains intubated and mechanically ventilated 6 hours after admission to the hospital for hematemesis. He has alcoholic cirrhosis and was brought to the emergency department 6 hours after he vomited a copious amount of blood. He has received 4 units of packed red blood cells. His current temperature is 37.4°C (99.3°F), pulse is 105/min, and blood pressure is 130/75 mm Hg. Examination shows peripheral muscle wasting. Cardiopulmonary examination shows no abnormalities. The abdomen is moderately distended and nontender. There is no fluid wave. His hematocrit is 29%. Placement of a nasogastric tube yields dark blood. Which of the following is the most appropriate next step in management?

- ☐ A) Radiolabeled erythrocyte scan
- ☐ B) Mesenteric arteriography
- ☐ C) Esophagogastroduodenoscopy
- ☐ D) Placement of a Sengstaken-Blakemore tube
- ☐ E) Surgical exploration of the abdomen

27. A 77-year-old man comes to the physician because of a 4-month history of difficulty swallowing solids and liquids associated with gurgling in his throat. During this period, he has been waking during the night because of coughing and has had bad breath. He has not had any other symptoms. He underwent appendectomy at the age of 25 years and cholecystectomy for gallstones at the age of 50 years. He currently takes no medications. Examination today shows no abnormalities except for halitosis. A barium swallow is shown. Which of the following is the most appropriate next step in management?

- ☐ A) Dietary counseling
- ☐ B) Therapy with a proton pump inhibitor
- ☐ C) Endoscopic dilatation of the esophagus
- ☐ D) 24-Hour monitoring of esophageal pH
- ☐ E) Diverticulectomy





28. A 16-month-old boy is brought to the emergency department by his parents after they noticed a small amount of bright red blood in his diaper 2 hours ago. The boy awakened twice overnight screaming as if in pain, and the parents first saw the blood after the second awakening. The patient ate well and was playing normally prior to going to bed. Ten days ago, he had an upper respiratory tract infection which resolved without treatment. On arrival, the child is sleeping and is difficult to arouse. Vital signs are within normal limits. Palpation of the abdomen discloses mild tenderness and a mobile, 4 × 3-cm mass. There is no distention, rebound, or guarding. Bowel sounds are normoactive. A small amount of blood and mucus is noted in the patient's diaper. The remainder of the examination shows no abnormalities. An abdominal x-ray series shows a nonspecific gas pattern. Which of the following is the most appropriate next step in diagnosis?

- ☐ A) Air enema with ultrasonography
- ☐ B) CT scan of the abdomen
- ☐ C) Laparotomy
- ☐ D) MRI of the abdomen
- ☐ E) Upper gastrointestinal series with small bowel follow-through

29. A 76-year-old woman is brought to the emergency department because of a 3-day history of increasing abdominal distention and left lower quadrant tenderness. The patient has a history of Parkinson disease and chronic constipation. Medications are carbidopa-levodopa and docusate and enemas as needed. Vital signs are within normal limits. The abdomen is distended and moderately tender to palpation over the left lower quadrant; no peritoneal tenderness, masses, or hernias are noted. Results of laboratory studies show a leukocyte count of $18,000/\text{mm}^3$. An abdominal x-ray is shown. Which of the following is the most likely diagnosis?

- ☐ A) Colon cancer
- ☐ B) Diverticular abscess
- ☐ C) Rectal prolapse
- ☐ D) Sigmoid volvulus
- ☐ E) Small bowel obstruction



30. A hospitalized 6-week-old girl with respiratory distress syndrome of the newborn has sudden episodes of apnea and temperature instability. Delivery was at 26 weeks' gestation. She has been exclusively fed with a cow milk-based formula. She appears pale and lethargic. Her temperature is 35.8°C (96.4°F), pulse is 160/min, respirations are 64/min, and blood pressure is 65/35 mm Hg. Capillary refill time is 6 seconds. Abdominal examination shows generalized distention and erythema around the umbilical stump. Bowel sounds are decreased. Laboratory studies show:

Leukocyte count	32,000/mm ³
Segmented neutrophils	10%
Bands	15%
Eosinophils	3%
Basophils	2%
Lymphocytes	70%
Platelet count	45,000/mm ³
Prothrombin time	30 sec (INR=2.5)
Partial thromboplastin time	70 sec

An abdominal x-ray is shown. Which of the following is the most likely diagnosis?

- ☐ A) Congenital megacolon (Hirschsprung disease)
- ☐ B) Hemorrhagic disease of the newborn
- ☐ C) Malrotation
- ☐ D) Necrotizing enterocolitis
- ☐ E) Omphalitis



31. A 45-year-old man comes to the office because of a 1-month history of rectal bleeding that occurs with bowel movements. The patient states the blood is mixed with the stool. He does not report any pain with bowel movements and has not experienced any weight loss or fatigue. Medical history is unremarkable, and he takes no medications. There is no family history of cancer. Vital signs are within normal limits. Results of abdominal and digital rectal examination show no abnormalities. Test of the stool for occult blood is negative. Which of the following is the most appropriate next step in diagnosis?

- ☐ A) Barium enema
- ☐ B) Capsule endoscopy
- ☐ C) Colonoscopy
- ☐ D) CT colonography
- ☐ E) Repeat test of stool for occult blood in 1 month



32. A 47-year-old homeless woman with alcoholism is brought to the emergency department after vomiting dark maroon fluid three times during the past 6 hours. Temperature is 37.0°C (98.6°F), pulse is 130/min, respirations are 24/min, and blood pressure is 80/40 mm Hg. There is dark, crusted vomitus around her mouth, and scleral icterus. Abdominal examination shows ascites. Mental status examination shows short-term memory loss; she is oriented to person and place but not to time. Test of the stool for occult blood is positive. Laboratory studies show:

Hemoglobin	7 g/dL
Leukocyte count	12,000/mm ³
Serum	
Total bilirubin	5 mg/dL
Direct	3.5 mg/dL

Administration of fluids is begun. Upper endoscopy shows esophageal varices with evidence of recent bleeding; no other abnormalities are noted. Which of the following is the most appropriate next step in management?

- ☐ A) Intravenous administration of ADH (vasopressin)
- ☐ B) Operative creation of a transjugular intrahepatic portosystemic shunt
- ☐ C) Placement of a Sengstaken-Blakemore tube
- ☐ D) Placement of a transjugular portosystemic stent
- ☐ E) Variceal banding

33. A previously healthy 37-year-old woman comes to the physician 6 hours after the onset of severe abdominal pain, fever, and chills. She takes no medications. Her temperature is 38.6°C (101.5°F). Abdominal examination shows severe right upper quadrant and epigastric tenderness. Laboratory studies show:

Hemoglobin	13 g/dL
Hematocrit	39%
Leukocyte count	18,000/mm ³ with a left shift
Serum	
Total bilirubin	1.2 mg/dL
Alkaline phosphatase	69 U/L
AST	19 U/L
ALT	15 U/L
Amylase	115 U/L

Which of the following is the most appropriate next step in diagnosis?

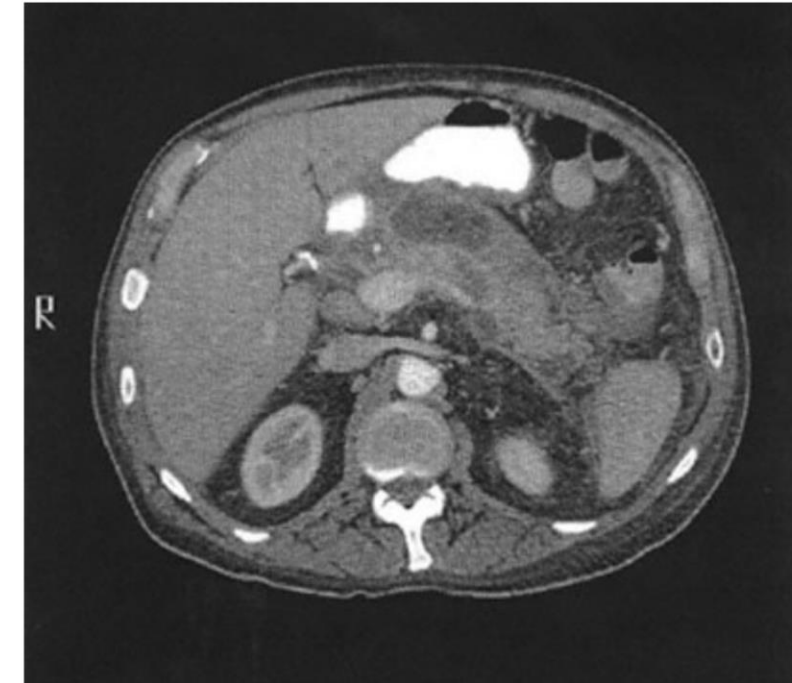
- ☐ A) Abdominal ultrasonography
- ☐ B) Upper gastrointestinal series
- ☐ C) Endoscopic retrograde cholangiopancreatography
- ☐ D) CT scan of the abdomen
- ☐ E) HIDA scan

34. A 67-year-old man comes to the emergency department because of the gradual onset of severe epigastric pain radiating to his back during the past 10 hours. He also has had nausea and vomiting during this period. He has a long-standing history of alcoholism and had a 2-day weekend drinking binge that ended 3 days ago. He has no history of serious illness and takes no medications. On arrival, he appears restless and leans forward slightly to relieve his pain. His temperature is 38°C (100.4°F), pulse is 120/min, respirations are 18/min and shallow, and blood pressure is 110/60 mm Hg. Abdominal examination shows severe epigastric tenderness to palpation. Laboratory studies show:

Hematocrit	38%
Leukocyte count	12,000/mm ³
Serum	
Urea nitrogen	45 mg/dL
Creatinine	1.0 mg/dL
Amylase	3215 U/L

A CT scan of the abdomen is shown. In addition to intravenous administration of fluids, which of the following is the most appropriate next step in management?

- ☐ A) Bowel rest
- ☐ B) Endoscopic retrograde cholangiopancreatography
- ☐ C) Percutaneous placement of a drain
- ☐ D) Peritoneal dialysis
- ☐ E) Surgical exploration of the abdomen



35. An 18-year-old man comes to the emergency department 48 hours after the sudden onset of right groin pain while lifting a heavy sofa. The pain has increased in intensity since onset, and he has vomited twice during the past hour. His temperature is 38.2°C (100.8°F), pulse is 100/min, respirations are 14/min, and blood pressure is 110/70 mm Hg. The right scrotum is edematous, slightly erythematous, acutely tender, and full. The abdomen is soft and mildly distended; bowel sounds are hyperactive. X-rays of the abdomen show a few air-fluid levels in the small bowel. Which of the following is the most appropriate next step in management?
- ☐ A) Application of an ice pack and elevation of the scrotum
 - ☐ B) Observation and broad-spectrum antibiotic therapy
 - ☐ C) Ultrasonography of the scrotum
 - ☐ D) Needle aspiration of the scrotum
 - ☐ E) Immediate surgical exploration

36. A hospitalized 77-year-old woman has had fever for the past 3 days. She underwent surgical repair of a perforated esophagus 4 weeks ago and has had a prolonged ileus. She received the diagnosis of pneumonia 16 days ago and has been intubated and mechanically ventilated since then. She is currently receiving broad-spectrum antibiotic therapy and parenteral nutrition. Her current temperature is 39°C (102.2°F). A central venous catheter is in place. Examination shows no rash. Heart sounds are normal. Blood cultures grow *Candida albicans*. In addition to intravenous fluconazole therapy, which of the following is the most appropriate next step in management?
- ☐ A) Renal ultrasonography
 - ☐ B) CT scan of the abdomen
 - ☐ C) Intravenous acyclovir therapy
 - ☐ D) Intravenous amphotericin B therapy
 - ☐ E) Removal of the central venous catheter



37. Forty-eight hours after undergoing mechanical aortic valve replacement, a 67-year-old man develops oliguria. His postoperative course was complicated by emergency reexploration 18 hours after the valve replacement because of bleeding. At that time, he received 12 units of packed red blood cells and was hypotensive for 3 hours. Anticoagulant therapy has been withheld because of concern about bleeding. Currently, his temperature is 37.2°C (99°F), pulse is 102/min, and blood pressure is 110/68 mm Hg. Pulmonary capillary wedge pressure is 12 mm Hg (N=5–16). His serum urea nitrogen concentration is 56 mg/dL, and serum creatinine concentration is 3.1 mg/dL. On admission, his serum urea nitrogen concentration was 12 mg/dL, and serum creatinine concentration was 0.9 mg/dL. Which of the following is the most likely cause of the oliguria?

- ☐ A) Acute transfusion reaction
- ☐ B) Acute tubular necrosis
- ☐ C) Hypoxia
- ☐ D) Pulmonary edema
- ☐ E) Renal artery embolus from the valve

38. A 27-year-old man is brought to the emergency department by paramedics 30 minutes after his motorcycle collided with an automobile. He was ejected from the seat on impact. He was wearing a helmet. He has severe pelvic pain. His pulse is 100/min, respirations are 28/min, and blood pressure is 120/70 mm Hg. The lower abdomen is tender to palpation; there is no guarding or rebound. Examination of the pelvis shows severe tenderness on compression of the iliac crest. There is blood at the urethral meatus, and ecchymosis over the scrotum and perineum. On digital rectal examination, the prostate cannot be palpated. Which of the following is the most likely diagnosis?

- ☐ A) Pelvic floor disruption
- ☐ B) Rectal laceration
- ☐ C) Ruptured bladder
- ☐ D) Testicular torsion
- ☐ E) Urethral transection

39. A 28-year-old woman comes to the office because of a 3-month history of a mass in the upper outer quadrant of her right breast. The patient says the mass fluctuates in size in relation to her menstrual cycle. She reports no pain or skin changes on or around the breast. Her family history is notable for a maternal aunt with a history of breast cancer diagnosed at age 45 years. Vital signs are within normal limits. Physical examination discloses a 1-cm mass that is mobile, well circumscribed, and nontender to palpation. No axillary or supraclavicular adenopathy is noted. The remainder of the examination discloses no abnormalities. Which of the following is the most appropriate next step in evaluation?
- ☐ A) Mammography
 - ☐ B) MRI of the breast
 - ☐ C) Needle-localized biopsy
 - ☐ D) Ultrasonography of the breast
 - ☐ E) No further evaluation is indicated



40. A 62-year-old woman has had a lump in her breast for 2 months. Her sister was diagnosed with breast cancer at the age of 65 years. Examination shows a $3 \times 2 \times 2$ -cm smooth mass in the upper outer quadrant of the right breast. There is no lymphadenopathy. Aspiration of the mass yields dark green fluid. No residual mass is palpable. Which of the following is the most likely diagnosis?

- ☐ A) Breast abscess
- ☐ B) Breast cancer
- ☐ C) Breast cyst
- ☐ D) Cystosarcoma phyllodes
- ☐ E) Fat necrosis



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41. A 68-year-old woman comes to the office for a routine examination. She has no history of serious illness and takes no medications. Vital signs are within normal limits. Examination of the breasts shows no abnormalities. Mammography of the right breast shows a 15-mm area of microcalcification; core needle biopsy results show intraductal breast carcinoma with two 1-mm areas of microinvasion. There are no abnormalities of the left breast. Which of the following is the most appropriate next step in management?

- ☐ A) Excisional biopsy only
- ☐ B) Excisional biopsy and radiation therapy
- ☐ C) Lumpectomy and sentinel node biopsy
- ☐ D) Mastectomy and axillary node dissection
- ☐ E) Tamoxifen therapy and repeat mammography in 6 months



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42. An afebrile 8-year-old boy has the sudden onset of pain in the right testis. Examination shows a swollen, tender, slightly erythematous right scrotum. Leukocyte count is normal. Which of the following is the most appropriate next step in management?

- ☐ A) Intravenous administration of antibiotics
- ☐ B) Application of warm compresses and reevaluation in 12 hours
- ☐ C) CT scan of the right scrotum
- ☐ D) Percutaneous needle aspiration and drainage
- ☐ E) Surgical exploration



43. A 57-year-old woman comes to the physician because of a nonhealing lesion on her foot for 8 weeks. She has type 1 diabetes mellitus treated with insulin. Examination shows a nontender 2-cm plantar ulcer over the head of the first metatarsal. Dorsalis pedis pulses are normal bilaterally. Which of the following is the most likely underlying cause of the ulcer?

- ☐ A) Arterial insufficiency
- ☐ B) Chronic osteomyelitis
- ☐ C) Lymphedema
- ☐ D) Peripheral neuropathy
- ☐ E) Venous stasis



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44. A 62-year-old woman comes to the office because she has had a nontender lump on the left side of her neck for 3 months. She has had no hoarseness or difficulty swallowing. Examination shows a firm, palpable mass in the superior pole of the left thyroid lobe; on ultrasound, it is irregular and measures 2 × 1.2 cm. There is no neck adenopathy. Which of the following is the most appropriate next step in diagnosis?
- ☐ A) CT scan of the neck
 - ☐ B) Direct laryngoscopy
 - ☐ C) Excisional biopsy
 - ☐ D) Fine-needle aspiration biopsy
 - ☐ E) ¹³¹I thyroid scan
 - ☐ F) Technetium 99m scan

45. A 29-year-old man with a 6-year history of end-stage renal disease undergoes uneventful renal transplantation. There is immediate diuresis of the kidney with urine output greater than 500 mL/h. Over the next several days, his serum creatinine concentration decreases to normal, and serum phosphorus concentration decreases to 1.5 mg/dL. The decrease in the serum phosphorus concentration is most likely because of which of the following?

- ☐ A) Hypercalcemia
- ☐ B) Hyperglycemia
- ☐ C) Hyperkalemia
- ☐ D) Hyponatremia
- ☐ E) Hyperparathyroidism

46. A 57-year-old man is being evaluated in the hospital 1 day after undergoing open reduction and internal fixation of a right hip fracture. He sustained a myocardial infarction intraoperatively. Coronary artery angiography showed two occluded arteries; angioplasty and stenting were performed. He has hypertension, congestive heart failure, hypercholesterolemia, obesity, and major depressive disorder. Preoperative medications included metoprolol, atorvastatin, and fluoxetine; all have been continued since admission. Currently, he is intubated and receiving morphine intravenously every 4 hours and 5% dextrose in 0.9% saline (100 mL/h). His temperature is 37.2°C (99°F), pulse is 88/min, respirations are 16/min, and blood pressure is 130/92 mm Hg. Examination shows a well-healing surgical incision on the right hip. There is sacral edema. Preoperatively, his serum sodium concentration was 134 mEq/L. Laboratory studies today show:

Serum	
Na ⁺	128 mEq/L
K ⁺	4.1 mEq/L
Cl ⁻	90 mEq/L
HCO ₃ ⁻	23 mEq/L
Urea nitrogen	11 mg/dL
Glucose	100 mg/dL
Creatinine	1 mg/dL
Urine	
Sodium	40 mEq/L
Osmolality	500 mOsmol/kg H ₂ O

Which of the following is the most likely cause of this patient's hyponatremia?

- ☐ A) Decreased total body sodium
- ☐ B) Ectopic ADH (vasopressin) release
- ☐ C) Excessive intravenous fluids
- ☐ D) Increased hypothalamic ADH release
- ☐ E) Shifting of water from the intracellular to extracellular compartment

47. A 37-year-old woman with stage IV breast cancer is admitted to the hospital because of lethargy and right thigh pain for the past 3 days. Her pulse is 100/min, respirations are 10/min, and blood pressure is 100/60 mm Hg. Physical examination shows dry mucous membranes. Mental status examination shows deficits in short-term memory. She is oriented to person but not to place or time. Her serum calcium concentration is 14.9 mg/dL. Administration of which of the following is the most appropriate next step in management?
- ☐ A) Calcitonin
 - ☐ B) Furosemide
 - ☐ C) Methylprednisolone
 - ☐ D) Plicamycin
 - ☐ E) 0.9% Saline
 - ☐ F) Sodium biphosphate



48. Five days after undergoing emergency surgery for a perforated appendix, a 67-year-old woman has abdominal pain and nausea; she has vomited twice during the past 30 minutes. She has coronary artery disease and sustained a myocardial infarction 2 years ago; she has been taking oral nitroglycerin since that time. She appears confused and agitated. Temperature is 38.8°C (101.8°F), pulse is 122/min, respirations are 23/min, and blood pressure is 88/50 mm Hg. There is no jugular venous distention. The lungs are clear to auscultation. Cardiac examination shows no abnormalities. There is mild distention of the abdomen with tenderness over the suprapubic area. Rectal examination shows mild tenderness on the right. This patient most likely has which of the following types of shock?

- ☐ A) Cardiogenic
- ☐ B) Extracardiac obstructive
- ☐ C) Hypovolemic
- ☐ D) Neurogenic
- ☐ E) Septic

49. A comatose 44-year-old woman is brought to the emergency department 30 minutes after sustaining a severe head injury in a motor vehicle collision. Her temperature is 36.2°C (97.1°F), pulse is 98/min, and blood pressure is 110/70 mm Hg. She is intubated and mechanically ventilated. The pupils are midrange and fixed, and gag and ciliary reflexes are absent. She does not respond to painful stimuli, and there are no spontaneous respirations. She has a depressed frontal skull fracture, and there is blood in the nares and right auditory canal. The family agrees to organ donation but requests proof of brain death other than the physical findings. Which of the following is the most effective test to determine brain death?
- ☐ A) Cerebral blood flow studies
 - ☐ B) CT scan of the head with contrast
 - ☐ C) Measurement of intracranial pressure
 - ☐ D) MRI of the brain
 - ☐ E) Skull series with tomography

50. An 18-year-old woman with ovarian germ cell cancer comes to the physician with her parents to discuss treatment options. Her condition has progressed despite chemotherapy, and the chemotherapy has caused multiple adverse effects. Her parents have researched a new experimental treatment protocol that would involve bone marrow transplantation but would only have a 5% 2-year survival likelihood. The daughter does not want to undergo transplantation because she says that she is ready to die. Her parents are distraught and want to do everything possible to improve her prognosis. Which of the following is the most appropriate next step in management?

- ☐ A) Ask the hospital ethics committee if the transplant should be performed
- ☐ B) Respect the wishes of the patient
- ☐ C) Obtain a court order for psychiatric assessment to determine the patient's decision-making capacity
- ☐ D) Proceed with the transplant at the parents' request